Propagation of Ornamental Plants
Vol. 00, № 0, 2000: 00-00

EFFECT OF IBA AND BA IN ROOTING AND GROWTH OF *CHRYSANTHEMUM* × *MORIFOLIUM* 'SNOWBALL' AXILLARY BUD CUTTINGS

Jaskirat Singh Josan*, Ranjit Singh, Madhu Bala, and Dhawan Shweta Machhindra

Department Floriculture and Landscaping, Punjab Agricultural University, Ferozepur Road (NH5), 141004 Ludhiana, Punjab, India, *E-mail: jsjosan055@gmail.com

Abstract

This study investigates an alternative multiplication method for *Chrysanthemum* × *morifolium* 'Snowball', circumventing the traditional use of terminal stem cuttings due to the limited availability of terminal shoots for commercial production in an *in vivo* environment. The objective was research the effect of indole-3-butyric acid (IBA) and N^6-benzyladenine (BA) on rooting and shoot formation of cuttings. The research was conducted in cycle 2023 and 2024. The highest rooting (93.0%) was observed in 75 mg l^{-1} IBA + 25 mg l^{-1} BA. Meanwhile, cuttings treated with 25 mg l^{-1} IBA exhibited the highest number of roots (20.7), the longest root length (16.3 cm), and the shortest time to shoot formation (19.1 days). The peak of peroxidase activity in the cuttings was observed on the 12th day with 75 mg l^{-1} IBA, reaching the highest level of 1.32 $\Delta A\ min^{-1}\ g^{-1}$ FW. The use of axillary bud cuttings proved to be a superior approach for the multiplication of 'Snowball', resulting in a four to fivefold increase in plant production compared to the traditional technique. The IBA 25mg l^{-1} proved to be the best treatment, outperforming comparable treatments in overall parameters.

Key words: development, explants, growth, propagation, shoot proliferation

PROOF

INTRODUCTION

The Chrysanthemum (*Chrysanthemum* × *morifolium* Ramat.), commonly referred to as "mum," is a notable ornamental plant with both historical significance and economic value, belonging to the Asteraceae family. Chrysanthemums are cultivated for a variety of purposes, including ornamental gardening, pot culture, and as cut flowers for bouquets and vase decoration. It has been demonstrated that *C. morifolium* possesses the capability to purify air by absorbing contaminants present in indoor environments (Irga et al. 2018). Chrysanthemums hold a prominent position in the floral market, ranking second only to roses in the global commercial flower trade (Kalia 2015). They are propagated through both sexual and asexual methods. The variation observed in seed-grown chrysanthemums, due to their highly cross-pollinated nature, polyploidy, and sporophytic self-incompatibility, necessitates vegetative propagation to maintain desirable traits (Karnawat et al 2021, Pu et al. 2021).

Plant growth regulators (PGRs) play a crucial role in enhancing the effectiveness of chrysanthemum propagation. Auxins, such as Indole-3-butyric acid (IBA), promote rooting, while cytokinins, including 6-benzyladenine (BA), are vital for shoot regeneration (Hartmann et al. 2017). Cytokinins are particularly effective in inducing adventitious bud formation in leaf cuttings, which is essential for successful propagation through the leaf with axillary bud methods (Singh and Chettri 2013). Despite advancements in propagation techniques, the propagation of chrysanthemums from standard-type stem cuttings is limited by the scarcity of stems. Consequently, the development of leaf nodal propagation methods is gaining importance, as leaf nodes are more abundant than terminal stems in chrysanthemum plants. Research on *Punica granatum* cv. Wonderful demonstrated that the rooting (100.0%) and the number of roots (15.2) were positively affected when the cutting was severed at an internode (Sarrou et al 2014). This method, when combined with the application of root-promoting PGRs has the potential to enhance propagation efficiency in commercial cultivation (Hartmann et al. 2017). According to Kumar et al. (2024) maintaining Reactive Oxygen Species (ROS) homeostasis is crucial for orchestrating root architecture which depend on auxin and its interactions with other signaling cascades. Further, the results shown that auxin treatment rescued the compromised root system development by restoring the ROS equilibrium (hydro-

Received: June 26, 2025

Accepted: August 10, 2025

gen peroxide) in stressed rice roots. Exogenous auxin application considerably increases Peroxidase (POD) synthesis rate ($p = 0.02$) and accelerates the timing of peak POD activity compared to control causing early emergence of adventitious roots (Zhang et al. 2017).

The aim of the study was to research the effect of IBA and BA on rooting and growth of axillary bud cuttings and how these PGRs regulate POD levels.

MATERIALS AND METHOD

Rooting of cuttings

The study was conducted at the Research Farm, Department of Floriculture and Landscaping, Punjab Agriculture University, Ludhiana, India, utilizing *Chrysanthemum morifolium* 'Snowball'. The research was carried out twice during the spring seasons of 2023 and 2024, employing axillary bud cuttings of uniform length (1.5-2 cm). These cuttings were used to overcome the limitations of conventional terminal cuttings, enabling a fivefold increase in planting material from a single plant. The basal parts of axillary bud cuttings were subjected to a quick dip (5 s) in alcohol solutions combination of IBA (0, 25, 50, and 75 mg l^{-1}) and BA (0, 25, and 50 mg l^{-1}) at varying concentrations. The cuttings in control treatment were treated only with alcohol. All cuttings were rooted in plug trays containing a substrate of sieved farmyard manure and sand (1 : 1, v : v). Relative air humidity of 80% and air temperature of 25°C were maintained by irrigation through automated misting system. Nozzles were activated 8 times with a duration of 6 min each during the day and misting remained off during the night. The light intensity was regulated using a 50% green shade net to ensure partial shading conducive to propagation. The experiment was designed as a completely randomized design. Each treatment was repeated 5 times and each repetition contained 25 cuttings. The rooted cuttings (%), number of roots, root length (cm), number of leaves, and shoot length (cm) were recorded after 50 days.

Peroxidase activity

Peroxidase (POD) activity was analyzed at four-day intervals using the guaiacol assay. Dissipated sampling was done with propagation of a different set of cuttings to observe peroxidase activity over time, and days to root initiation by removing the cuttings from the substratum in plug trays every two days. POD enzyme activity was determined by monitoring the change in absorbance ($\Delta A\ min^{-1}g^{-1}$ FW) in a reaction mixture. The extraction buffer consisted of 0.1 M phosphate buffer (pH 6.5) with 1% polyvinylpyrrolidone (PVP), 1 mM EDTA, and 10 mM β-mercaptoethanol. The buffer was prepared by mixing solutions NaH_2PO_4, monobasic phosphate (34 ml) and Na_2HPO_4, dibasic phosphate (16 ml), then diluting with distilled water to a final volume of 100 ml. For PVP, 1 g of polyvinylpyrrolidone was dissolved in 80 ml of phosphate buffer, heated, cooled, and made up to 100 ml. EDTA was prepared by dissolving 0.03 g in the same buffer with 1% PVP, then making up to 100 ml. Mercaptoethanol (139.2 µl) was added to a 200 ml solution of phosphate buffer, PVP, and EDTA. A 0.05 M guaiacol solution was prepared by dissolving 0.554 ml of guaiacol in 100 ml of 0.1 M phosphate buffer. A 0.8 M H_2O_2 solution was freshly prepared by adding 0.83 µl of 30% H_2O_2 to 10 ml distilled water. To extract enzymes, 0.1 g of tissue from the basal node region was crushed with extraction buffer in an ice bath, followed by centrifugation at 10,000 rpm for 20 min at 4°C. The supernatant was used for peroxidase activity measurement. The assay mixture consisted of 500 µl of enzyme extract and 25 ml of phosphate buffer, with 100 µl of 0.8 M H_2O_2 to initiate the reaction. The absorbance change was recorded at 470 nm every 30 s for 3 min. The enzyme activity was expressed as $\Delta A\ min^{-1}g^{-1}$, and specific activity was calculated by normalizing the enzyme activity to the protein concentration in the extract.

RESULTS AND DISCUSSION

Rooting of cuttings

The highest rooting rate of cuttings of 93.0% was observed with 75 mg l^{-1} IBA + 25 mg l^{-1} BA, which was 27% better as compared to control. Optimizing IBA concentrations for enhanced rooting in chrysanthemum propagation affirmed the general trend observed in the current study, revealing that IBA concentrations within a specific range significantly enhanced rooting percentages in chrysanthemum cuttings (Ullaha 2023). Auxin applied exogenously significantly surge endogenous levels of IAA (Zhou et al. 2023). Moreover, the similar study shows earlier and more effective adventitious root initiation is strongly associated with increased endogenous IAA, highlighting the vital role of internal auxin signalling. The treatment of *Portulaca umbraticola* Kunth stem cuttings with BA in combination with NAA proved increase in the percentage of their rooting and BA at 0.5 g dm^{-3} concentration doubled the root number (Wróblewska and Bąbelewski 2010). The shortest root initiation period of 14.3 days was observed in cuttings treated with 75 mg l^{-1} IBA. However, an increase in IBA concentration was associated with an accelerated root initiation process, as evidenced by the results for IBA at 50 mg l^{-1} (16.3 days) and IBA at 25 mg l^{-1} (17.2 days) (Table 1). This suggests that exogenous auxin treatment promotes the recruitment of carbohydrates in the shoot, thereby enhancing sugar availability at the site of root primordia formation, which serves as a significant carbon source. These sugars are utilized to provide the energy necessary for cell division and differentiation, initiating root formation (Agulló-Antón et al. 2011).

Conversely, higher concentrations of BA in treatments delayed root initiation, with untreated cuttings requiring the longest time (20.6 days) for root initiation (Table 1). These findings are consistent with the study by Chang et al. (2015), which highlights the role of BA in delaying root initiation. This delay is attributed to the spatiotemporal expression pattern of cytokinin synthesis genes, Isopentenyl Transferase (IPT) and Lonely Guy (LOG), which generate a cytokinin source in developing primordia, forming an inhibitory field that prevents lateral root initiation in adjacent cells. Furthermore, IPT3, IPT5, and IPT7 inhibit the initiation of new lateral roots by promoting cytokinin production near existing lateral root primordia. This inhibitory effect can be counteracted by high concentrations of auxin (Du and Scheres 2018). These buds, being meristematic, stimulate rooting by synthesizing and supplying growth substances necessary for root initiation (Ghosh et al. 2022).

The maximum number of roots (20.7) was observed in 25 mg l^{-1} IBA, which was followed by to 50 mg l^{-1} IBA, with 17.6 roots (Table 1). This finding aligns with the study by Poulsen and Andersen (1980), which reported an increase in root number with increasing distance from the apex in *Hedera helix* L. Optimal IBA concentration results in reduced time required for callus formation, further enhancing cambium dedifferentiation, producing numerous cells that will differentiate to form root cells. Amyloplast levels naturally decline during rooting, and optimal IBA accelerates the disappearance of amyloplasts (Singh et al. 2014). The lowest root number (13.2), was observed in 75 mg l^{-1} IBA combined with 50 mg l^{-1} BA, which was comparable to the control (14.6) and 75 mg l^{-1} IBA combined with 25 mg l^{-1} BA (13.5). The regression in root number indicates a toxic effect when IBA concentration exceeds the optimum in relation to high BA concentration, leading to a reduction in the number of roots (Table 1, Fig. 1). Ahmed et al. (2025) affirmed a similar observation in *Crocus savitus* L.where the optimal concentration of auxin application resulted reduction in rooting. The effect of auxin on rooting is promotive at low concentrations and inhibitory at high concentrations (De Klerk et al 1997).

The longest roots (16.3 cm), were observed after treatment with 25 mg l^{-1} IBA, demonstrating significantly better convergence with other treatments (Table 1, Fig. 1). This suggests that IBA application at this concentration enhances carbohydrate hydrolysis, metabolite accumulation, protein synthesis, and auxin-induced cell division and expansion, as noted in previous studies (Strydom and Hartmann 1960, Janakiram et al. 2006). Although the primary function of auxin is to regulate root elongation, the shortest root length of 9.2 cm was observed after treatment with 75 mg l^{-1} IBA, which performed worse than the untreated cuttings (Table 1). This indicates that the higher concentration of IBA may result in toxicity, leading to reduced root length (Joel et al. 2023).

Shoot proliferation and elongation

Early shoot formation, indicated by the emergence of axillary buds, was observed most rapidly in treatments with 25 mg l^{-1} IBA and 75 mg l^{-1} IBA, approximately 19.1 days, respectively. Treatments with 25 mg l^{-1} IBA + 25 mg l^{-1} BA, 50 mg l^{-1} IBA, and 75 mg l^{-1} IBA + 25 mg l^{-1} BA were comparable to 25 mg l^{-1} IBA. The longest shoot formation duration (23.1 days) was recorded in the control treatment (Table 2). The acceleration in early sprouting may be attributed to IBA's

Table 1. Effect of IBA and BA concentrations on rooting of cuttings (Pooled analysis year 2023 and 2024).

Treatment	Rooted cuttings (%)		Days to root initiation		Number of roots		Root length (cm)	
	2023	2024	2023	2024	2023	2024	2023	2024
Control	73.0 ± 0.7 d	72.6 ± 1.5 d	20.2 ± 0.7 ab	21.1 ± 0.7 a	15.2 ± 1.2 cde	14.0 ± 0.4 ef	10.0 ± 0.4 fg	9.9 ± 0.4 ef
25 mg l^{-1} IBA	90.4 ± 1.2 bc	90.4 ± 1.1 abc	17.2 ± 0.4 de	17.3 ± 0.3 ef	20.8 ± 0.5 a	20.5 ± 0.6 a	15.9 ± 0.7 a	16.7 ± 0.1 a
25 mg l^{-1} IBA + 25 mg l^{-1} BA	90.6 ± 1.1 bc	90.0 ± 0.7 abc	18.3 ± 0.3 cd	18.5 ± 0.4 d	17.0 ± 0.5 bc	17.1 ± 0.5 bc	12.4 ± 0.5 bcd	12.9 ± 0.6 b
25 mg l^{-1} IBA + 50 mg l^{-1} BA	92.0 ± 1.6 ab	92.2 ± 1.0 a	20.3 ± 0.3 a	20.4 ± 0.3 ab	15.6 ± 0.9 bcd	15.7 ± 0.6 cd	10.7 ± 0.6 def	10.4 ± 0.5 e
50 mg l^{-1} IBA	88.4 ± 1.2 c	89.8 ± 1.0 bc	16.2 ± 0.5 e	16.5 ± 0.5 f	17.8 ± 0.8 b	17.3 ± 0.6 b	11.5 ± 0.4 cd	11.5 ± 0.5 cd
50 mg l^{-1} IBA + 25 mg l^{-1} BA	88.6 ± 0.9 c	90.2 ± 0.8 abc	19.0 ± 0.2 bc	18.9 ± 0.3 cd	17.1 ± 1.0 bc	17.3 ± 0.4 b	11.3 ± 0.4 cd	11.9 ± 0.3 bcd
50 mg l^{-1} IBA + 50 mg l^{-1} BA	87.8 ± 0.7 c	87.6 ± 0.5 c	19.8 ± 0.4 ab	19.9 ± 0.4 bc	16.2 ± 0.8 bc	16.6 ± 0.8 bcd	11.8 ± 0.2 bcd	11.2 ± 0.4 cde
75 mg l^{-1} IBA	92.2 ± 0.8 ab	90.6 ± 0.3 ab	14.3 ± 0.4 f	14.3 ± 0.3g	15.5 ± 1.4 bcde	15.2 ± 0.3 de	9.4 ± 0.4 fgh	9.0 ± 0.2 f
75 mg l^{-1} IBA + 25 mg l^{-1} BA	94.2 ± 0.7 a	91.8 ± 0.5 a	17.3 ± 0.4 de	17.2 ± 0.3 f	13.6 ± 0.6 de	13.4 ± 0.5 f	11.0 ± 0.5 de	11.6 ± 0.6 bcd
75 mg l^{-1} IBA + 50 mg l^{-1} BA	92.0 ± 1.1 ab	91.2 ± 0.5 ab	18.5 ± 0.5 c	18.4 ± 0.5 de	13.1 ± 0.7 e	13.3 ± 0.4 f	13.1 ± 0.4 b	12.9 ± 0.4 b

Means ± standard errors within a column followed by the same letter are not significantly different according Tukey's HSD multiple range test at $p \leq 0.05$.

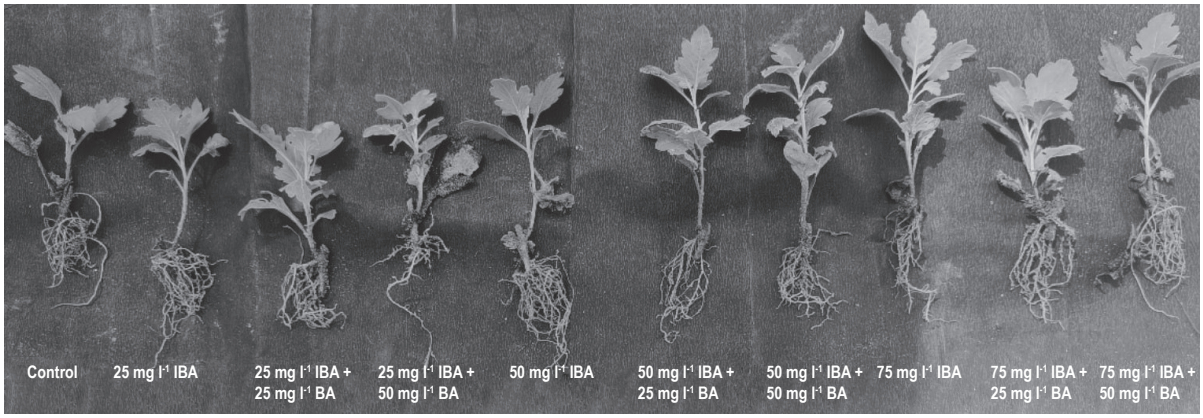


Fig. 1. Effect of IBA and BA on the rooting and shoot formation of chrysanthemum 'Snowball' axillary bud cuttings.

enhanced ability to hydrolyze reserve food materials and regulate endogenous auxin levels by increasing IAA oxidase activity (Shukla et al. 2010). Similar early axillary shoot growth was optimally achieved *Rosa hybrida* L. by treating cuttings with IBA concentrations ranging from 10^{-4} to 10^{-3} M, while the highest concentration of IBA 10^{-1} M delayed axillary shoot growth by more than a week compared to lower concentrations (Bredmose et al. 2004). Higher concentrations of IBA combined with BA influence shoot formation in *Fraxinus nigra* Marsh., as BA promotes cell differentiation and shoot proliferation (Lee and Pijut 2017). The extended time to shoot formation in the control treatment may be due to the absence of exogenous PGRs that catalyze and enhance growth by improving the plant's internal efficiency.

The highest number of leaves 9.5, was observed after treatment with 50 mg l^{-1} IBA + 50 mg l^{-1} BA, cuttings treated with 25 mg l^{-1} IBA + 25 mg l^{-1} BA and 50 mg l^{-1} IBA + 25 mg l^{-1} BA shows similar 9.2 leaves respectively, while 75 mg l^{-1} IBA + 25 mg l^{-1} BA showing 9.1 leaves, and 75 mg l^{-1} IBA + 50 mg l^{-1} BA showing 9.0 leaves, all of which were comparable to the best-performing treatment. These increases in leaf number were significant compared to the control treatment (Table 1, Fig. 1), highlighting the role of BA in enhancing early and maximum leaf development through early shoot proliferation, while IBA supports root structuring and enhancement. Cytokinin (CK) and increased auxin transport support CK-mediated bud proliferation which demonstrates that CK contributes to the branching phenotype observed in the ARR1 mutant by driving the accumulation of PIN3, PIN4, and PIN7 on the plasma membrane (Waldie and Leyser 2018).

The most significant result in shoot length (10.0 cm) was observed in the treatment with 50 mg l^{-1} IBA + 25 mg l^{-1} BA. Comparable results were obtained with 25 mg l^{-1} IBA + 25 mg l^{-1} BA, which produced a shoot length of 9.5 cm, and 75 mg l^{-1} IBA + 50 mg l^{-1} BA, which resulted in a shoot length of 9.7 cm. The untreated cuttings demonstrated poor performance, yielding the shortest shoot length after 50 days of the beginning of experiment. Each treatment involving BA underscored its critical role in shoot development. Our findings align with the research conducted by Al-

Table 2. Effect of IBA and BA concentrations on shoot proliferation in cuttings (Pooled analysis year 2023 and 2024).

Treatment	Days to shoot formation		Number of leaves		Shoot length (cm)	
	2023	2024	2023	2024	2023	2024
Control	22.7 ± 0.6 a	23.5 ± 0.5 a	5.6 + 0.1 c	5.4 + 0.2 d	6.7 + 0.2 d	6.3 + 0.1 f
25 mg l^{-1} IBA	19.3 ± 0.4 d	19.0 ± 0.3 d	8.6 ± 0.6 ab	8.4 ± 0.4 b	9.0 ± 0.4 abc	9.1 ± 0.4 bcd
25 mg l^{-1} IBA + 25 mg l^{-1} BA	19.3 ± 0.4 d	19.0 ± 0.4 d	9.4 ± 0.7 a	9.0 ± 0.5 ab	9.5 ± 0.3 ab	9.6 ± 0.3 abc
25 mg l^{-1} IBA + 50 mg l^{-1} BA	21.6 ± 0.3 ab	21.5 ± 0.3 b	8.6 ± 0.2 ab	8.4 ± 0.2 b	8.5 ± 0.3 bc	8.6 ± 0.3 de
50 mg l^{-1} IBA	19.2 ± 0.6 d	19.2 ± 0.5 d	7.0 ± 0.5 a	8.1 ± 0.4 ab	8.2 ± 0.2 c	7.5 ± 0.5 e
50 mg l^{-1} IBA + 25 mg l^{-1} BA	21.9 ± 0.5 ab	21.3 ± 0.4 b	9.4 ± 0.4 bc	9.0 ± 0.4 c	10.1 ± 0.5 a	10.0 ± 0.5 a
50 mg l^{-1} IBA + 50 mg l^{-1} BA	20.2 ± 0.4 cd	20.0 ± 0.4 cd	9.2 ± 0.5 a	9.7 ± 0.4 a	8.9 ± 0.8 abc	9.0 ± 0.4 bcd
75 mg l^{-1} IBA	19.1 ± 0.4 d	19.0 ± 0.4 d	8.9 ± 0.4 a	8.2 ± 0.4 b	8.9 ± 0.4 abc	8.9 ± 0.3 bcd
75 mg l^{-1} IBA + 25 mg l^{-1} BA	19.3 ± 0.5 d	19.1 ± 0.5 d	9.2 ± 0.3 a	8.9 ± 0.3 ab	9.2 ± 0.4 abc	9.3 ± 0.4 bcd
75 mg l^{-1} IBA + 50 mg l^{-1} BA	20.6 ± 0.4 bc	20.4 ± 0.4 bc	8.9 ± 0.6 ab	9.1 ± 0.5 ab	9.7 ± 0.5 ab	9.8 ± 0.3 ab

Means ± standard errors within a column followed by the same letter are not significantly different according Tukey's HSD multiple range test at $p \leq 0.05$.

Jubori and Al-Amery (2022) on *Citrus aurantifolia* (Christm.) Swingle propagation under *in vitro* conditions, which indicates that increased concentrations of cytokinins (CKs) within the branches diminish the function of accumulated auxins, leading to shorter shoots. Auxins and CKs are pivotal in regulating shoot elongation through a complex interplay of cellular and molecular mechanisms. Auxins, particularly IAA, are essential for shoot cell elongation by facilitating cell wall loosening, a process driven by turgor-induced expansion. This mechanism involves the activation of plasma membrane proton pumps, resulting in cell wall acidification and the subsequent activation of enzymes such as EXPANSINS, which break down bonds in the cell wall matrix to enable elongation (Majda and Robert 2018). Auxins regulate the expression of key genes associated with cell elongation and division, ensuring synchronized growth at the molecular level (Key 1989, Campanoni and Nick 2005). Conversely, CKs primarily promote shoot elongation by stimulating cell division within the shoot apical meristem, thereby establishing the structural basis for elongation (Murray et al. 2012, Yang et al. 2021). CKs regulate the expression of meristematic regulatory genes such as WUSCHEL, maintaining a robust stem cell population to support continuous shoot growth (Wang et al. 2017, Snipes et al. 2018). Additionally, CKs enhance nutrient redistribution to actively elongating shoots, indirectly facilitating growth (Prasad 2022).

Peroxidase activity

The activity of the peroxidase enzyme (POD) was initially minimal prior to the propagation of cuttings, with a notable increase observed after the fourth day. This initial rise in activity persisted until the twelfth day in the treatment with 75 mg l^{-1} IBA, reaching its peak level (1.324 ΔA $min^{-1}g^{-1}$ FW) just before root initiation. A gradual decline in peroxidase activity was observed following root initiation, a trend consistent across different treatments. The slowest peroxidase activity was recorded in the IBA 25 mg l^{-1} + IBA 50 mg l^{-1} treatment (1.066 ΔA $min^{-1}g^{-1}$ FW) at 20 days of propagation, with the untreated cuttings exhibiting comparable enzymatic activity (1.088 ΔA$min^{-1}g^{-1}$ FW) (Table 3). The concentration of IBA in the treatment also effected the maximum level of peroxidase enzyme activity, indicating that higher concentrations of auxin enhance peroxidase activity and expedite the process to reach the maximum level, thereby reducing the time to root initiation and directly as well as indirectly enhance the growth of plants (Table 3). Similar findings have been reported, suggesting that auxin concentrations exceeding the optimal range for rooting *Rhododendron* 'Babites Baltais' leaf bud cuttings contributed to the increase in peroxidase activity (Megre et al. 2011). Auxins and other PGRs induce metabolic changes in the rooting zones of *Tectona grandis* L. cuttings, which either promote or inhibit adventitious root growth (Husen and Pal 2007). Exogenous PGRs functioning depends highly on its transport pathway along with its concentration for better adventitious root formation process (Kunc et al. 2025). PODs help generate hydroxyl radicals, which promote cell wall loosening and elongation while maintaining the right ROS balance and extensin cross-linking in growing cells (notably root hairs), enabling faster, sustained elongation (Schopfer 2001, Pacheco et al. 2022). But excess POD activity depletes IAA and impairs growth (Stepanova and Alonso 2016). Maturity of cuttings also determines the requirement of different auxin levels (Osterc and Štampar 2011).

Table 3. Peroxidase activity (ΔA/min g^{-1}) over time with varying concentration of IBA and BA treatments on cuttings.

Treatments	Day 0	Day 4	Day 8	Day 12	Day 16	Day 20	Day 24
Control	0.01 + 0.001 a	0.05 + 0.004 e	0.17 + 0.007 e	0.38 + 0.006 g	0.69 + 0.003 d	1.09 + 0.023 a	0.30 + 0.011 a
25 mg l-1 IBA	0.01 + 0.001 a	0.12 + 0.006 c	0.42 + 0.017 c	0.61 + 0.007 e	1.04 + 0.005 b	0.28 + 0.007 cd	0.04 + 0.002 bc
25 mg l-1 IBA + 25 mg l-1 BA	0.01 + 0.001 a	0.12 + 0.002 c	0.41 + 0.015 c	0.52 + 0.005 f	1.01 + 0.035 bc	0.38 + 0.005 b	0.06 + 0.002 b
IBA 25 mg l-1 + BA 50 mg l-1	0.01 + 0.001 a	0.06 + 0.002 e	0.28 + 0.006 d	0.45 + 0.004 fg	0.72 + 0.02 d	1.07 + 0.028 a	0.29 + 0.01 a
50 mg l-1 IBA	0.01 + 0.001 a	0.08 + 0.006 b	0.55 + 0.004 b	0.78 + 0.006 bc	1.15 + 0.024 a	0.26 + 0.005 d	0.03 + 0.002 cd
50 mg l-1 IBA + 25 mg l-1 BA	0.01 + 0.001 a	0.13 + 0.005 c	0.56 + 0.017 b	0.74 + 0.004 cd	0.98 + 0.006 bc	0.32 + 0.003 bcd	0.06 + 0.002 b
50 mg l-1 IBA + 50 mg l-1 BA	0.01 + 0.001 a	0.09 + 0.009 d	0.52 + 0.005 b	0.68 + 0.007 de	0.96 + 0.007 c	0.33 + 0.008 bc	0.06 + 0.001 b
75 mg l-1 IBA	0.01 + 0.001 a	0.23 + 0.006 a	0.66 + 0.011 a	1.32 + 0.025 a	0.45 + 0.01 e	0.13 + 0.005 e	0.01 + 0.001 d
75 mg l-1 IBA + 25 mg l-1 BA	0.01 + 0.001 a	0.20 + 0.003 b	0.53 + 0.006 b	0.84 + 0.033 b	1.19 + 0.005 a	0.37 + 0.005 b	0.01 + 0.001 d
75 mg l-1 IBA + 50 mg l-1 BA	0.01 + 0.001 a	0.18 + 0.002 b	0.53 + 0.01 b	0.82 + 0.016 b	1.21 + 0.001 a	0.32 + 0.004 bcd	0.01 + 0.001 d

Means ± standard errors within a column followed by the same letter are not significantly different according Tukey's HSD multiple range test at $p \leq 0.05$.

Correlation matrix of morphological parameters with peroxidase activity

The correlogram (Fig. 2) depicts the interrelationships among morphological parameters with peroxidase activity. Positive correlations, indicated by greenish hues, include a strong association between "days to root initiation" and "peroxidase activity" (r = 0.85), signifying a close link between developmental timing and enzymatic activity for early rooting. Negative correlations, represented in brownish tones, include the relationship of "rooting percentage" and "number of leaves" with "peroxidase activity" (r = −0.49; r = −0.46), suggesting that the regular ability to form roots and leaf production may be constrained by POD activity. Moderate correlations, such as between "days to shoot formation" and "peroxidase activity" (r = 0.64), underscore interdependencies between shoot initiation and POD enzymatic activity that may indirectly be crucial for shoot development. The inclusion of numeric correlation values enhances interpretability, providing valuable insights into the complex interactions among morphological parameters and peroxidase activity, which can aid in optimizing plant growth and physiological outcomes.

Acknowledgments: We would like to thank Department of Floriculture and Landscaping for providing the necessary facilities.

REFERENCES

Agulló-Antón M. Á., Sánchez-Bravo J., Acosta M., Druege U. (2011). Auxins or sugars: what makes the difference in the adventitious rooting of stored carnation cuttings? Journal of Plant Growth Regulation, 30: 100-113.

Ahmed Z. S., Salim A. M., Al-Dur E. T. H., Al-

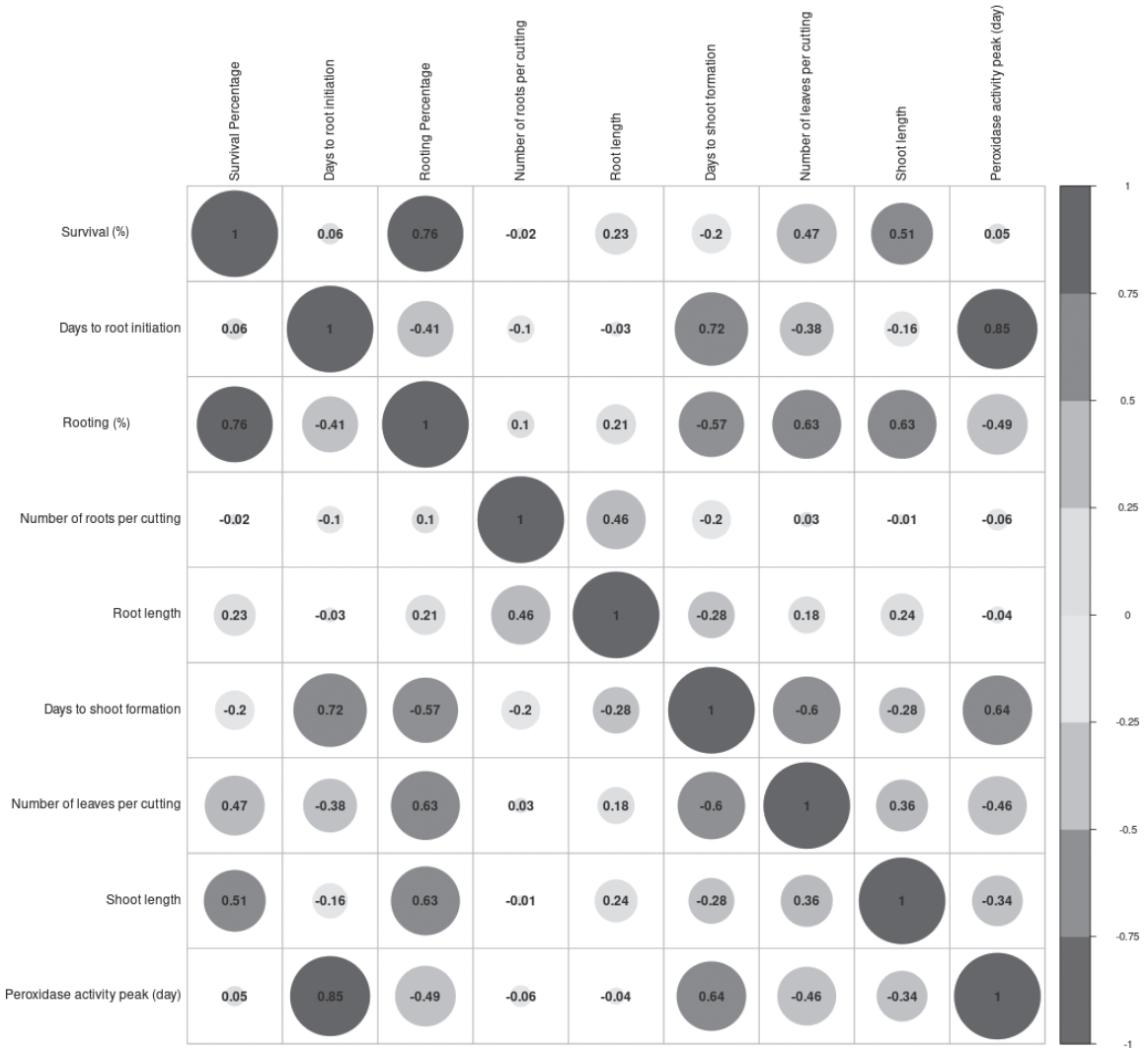


Fig. 2. Correlation among morphological traits and peroxidase activity. Spearman's rank correlation coefficients were researched for significance at $p < 0.05$, with $|r| \geq 0.20$ being considered significantly different from zero.

Chalabi A. T. M., Sanam M. A., Dargiri S. A., Meftahizade H. (2025). Optimizing micropropagation and microcorm induction in saffron (*Crocus sativus* L.) using PGRs (NAA and BAP) and elicitor salicylic acid. BMC Plant Biology, 25: 1-13.

Al-Jubori M. T., Al-Amery L. K. J. (2022). Effect of cytokinins BA, Kin, auxin IAA and IBA on propagating (*Citrus aurantiifolia*) *in-vitro*. International Journal of Agricultural and Statistical Sciences, 18: 2033-2040.

Bredmose N., Kristiansen K., Nielsen B. (2004). Propagation temperature, PPFD, auxin treatment, cutting size and cutting position affect root formation, axillary bud growth and shoot development in miniature rose (*Rosa hybrida* L.) plants and alter homogeneity. Journal of Horticultural Science & Biotechnology, 79: 458-465.

Campanoni P., Nick P. (2005). Auxin-dependent cell division and cell elongation 1-Naphthaleneacetic acid and 2,4-dichlorophenoxyacetic acid activate different pathways. Plant Physiology, 137: 939-948.

Chang L., Ramireddy E., Schmülling T. (2015). Cytokinin as a positional cue regulating lateral root spacing in *Arabidopsis*. Journal of Experimental Botany, 66: 4759-4768.

De Klerk G.-J., Brugge J. T., Marinova S. (1997). Effectiveness of indoleacetic acid, indolebutyric acid and naphthaleneacetic acid during adventitious root formation *in vitro* in *Malus* 'Jork 9'. Plant Cell, Tissue and Organ Culture, 49: 39-44.

Du Y., Scheres B. (2018). Lateral root formation and the multiple roles of auxin. Journal of Experimental Botany, 69: 155-167.

Ghosh S. N., Tarai R. K., Ahlawat T. R. (2022). Plant growth regulators in tropical and sub-tropical fruit crops. Narendra Publishing House, Delhi: 9-10.

Hartmann H. T., Kester D. E., Davies F. T., Geneve R. L. (2017). Plant propagation principles and practices. Pearson India Education Services Private Limited, Banglore: 295-360.

Husen A., Pal M. (2007). Metabolic changes during adventitious root primordium development in *Tectona grandis* Linn. f. (teak) cuttings as affected by age of donor plants and auxin (IBA and NAA) treatment. New Forests, 33: 309-323.

Irga P. J., Pettit T. J., Torpy F. R. (2018). The phytoremediation of indoor air pollution: a review on the technology development from the potted plant through to functional green wall biofilters. Reviews in Environmental Science and Biotechnology, 17: 395-415.

Janakiram T., Mahantesh M. I., Prabhakar B. S. (2006). Standardization of agro techniques for production of chrysanthemum under low-cost polyhouse. Acta Horticulturae, 710: 321-327.

Joel P. G., Bala M., Kaur A., Dubey R. K. (2023). Efficacy of rooting hormones on propagation and rooting potential of Japanese chrysanthemums (*Chrysanthemum morifolium* Ramat.). Rhizosphere, 27: 1-4.

Kalia R. (2015). Effect of different concentrations of auxins on the regeneration of *Chrysanthemum morifolium* plantlets. International Journal of Technology Research, 3: 106-107.

Karnawat M., Trivedi S. K., Meena R. K., Nagar D. (2021). A review on haploid and double haploids in ornamental plants. Current Research in Agriculture and Farming, 2: 1-7.

Key J. L. (1989). Modulation of gene expression by auxin. BioEssays 11: 52-58.

Kumar A., Verma K., Kashyap R., Joshi V. J., Sircar D., Yadav S. R. (2024). Auxin-responsive ROS homeostasis genes display dynamic expression pattern during rice crown root primordia morphogenesis. Plant Physiology and Biochemistry, 206: Article 108307.

Kunc P., Medič A., Hudina M., Veberič R., Osterc G. (2025). Physiological age of stock plants determines phytohormonal changes in leafy cuttings of *Prunus subhirtella* 'Autumnalis'. Journal of Plant Growth Regulation, 44: 721-730.

Lee J. H., Pijut P. M. (2017). Adventitious shoot regeneration from *in vitro* leaf explants of *Fraxinus nigra*. Plant Cell, Tissue and Organ Culture, 130: 335-343.

Majda M., Robert S. (2018). The role of auxin in cell wall expansion. International Journal of Molecular Sciences, 19: Article 951.

Megre D., Dokane K., Kondratovic U. (2011). Can changes in starch content and peroxidase activity be used as rooting phase markers for rhododendron leaf bud cuttings? Acta Biologica Cracoviensia, Series Botanica, 53: 74-79.

Murray J. A., Jones A., Godin C., Traas J. (2012). Systems analysis of shoot apical meristem growth and development: integrating hormonal and mechanical signaling. The Plant Cell, 24: 3907-3919.

Osterc G., Štampar F. (2011). Differences in endo/exogenous auxin profile in cuttings of different physiological ages. Journal of Plant Physiology, 168: 2088-2092.

Pacheco J. M., Ranocha P., Kasulin L., Fusari C. M., Servi L., Aptekmann A. A., Gabarain V. B., Peralta J. M., Borassi C., Marzol E., Rodríguez-Garcia D. R. (2022). Apoplastic class III peroxidases PRX62 and PRX69 promote *Arabidopsis* root hair growth at low temperature. Nature Communications, 13: Article 1310.

Poulsen A., Andersen A. S. (1980). Propagation of *Hedera helix*: influence of irradiance to stock plants, length of internode and topophysis of cutting. Physiologia Plantarum, 49: 359-365.

Prasad R. (2022). Cytokinin and its key role to en-

rich the plant nutrients and growth under adverse conditions-an update. Frontiers in Genetics, 13: Article 883924.

Pu Y., Huo R., Lin Q., Wang F., Chun X., Huang H., Dai S. (2021). Investigation and screening of *Chrysanthemum* resources to identify self-compatible mutants. Scientia Horticulturae, 281: 1-15.

Sarrou E., Therios I., Theriou K. D. (2014). Melatonin and other factors that promote rooting and sprouting of shoot cuttings in *Punica granatum* cv. Wonderful. Turkish Journal of Botany, 38: 293-301.

Schopfer P. (2001). Hydroxyl radical-induced cell-wall loosening *in vitro* and *in vivo*: implications for the control of elongation growth. The Plant Journal, 28: 679-688.

Shukla H. S., Tripathi V. K., Awasthi R. D., Tripathi A. K. (2010). Effect of IBA, PHB and boron on rooting and shoot growth of hard wood stem cuttings of peach. International Journal of Applied Agricultural Research, 5: 467-473.

Singh K. K., Choudhary T., Prabhat K., Rawat J. M. S. (2014). Effect of IBA for inducing rooting in stem cuttings of *Duranta* golden. HortFlora Research Spectrum, 3: 77-80.

Singh P., Chettri R. (2013). A new propagation method for rapid multiplication of *Chrysanthemum* under *in-vivo* conditions. International Journal of Conservation Science, 4: Article 95.

Snipes S. A., Rodriguez K., DeVries A. E., Miyawaki K. N., Perales M., Xie M., Reddy G. V. (2018). Cytokinin stabilizes *WUSCHEL* by acting on the protein domains required for nuclear enrichment and transcription. Public Library of Science Genetics, 14: Article 1007351.

Stepanova A. N., Alonso J. M. (2016). Auxin catabolism unplugged: role of IAA oxidation in auxin homeostasis. Proceedings of the National Academy of Sciences, 113: 10742-10744.

Strydom D. K., Hartmann H. T. (1960). Effect of indole butyric acid on respiration and nitrogen metabolism in Marianna 2624 plum softwood stem cuttings. American Society for Horticultural Science, 76: 124-133.

Ullaha R., Chachar M., Rasoolb A., Khanc A., Ahmeda Z., Ahmeda S., Hakroa R. A. (2023). Assessing the impact of indole-3-butyric acid (IBA) on rooting efficiency in carnation (*Dianthus caryophyllus* L.) propagation under protected conditions. Agricultural Extension and Development in Countries, 1: 43-45.

Waldie T., Leyser O. (2018). Cytokinin targets auxin transport to promote shoot branching. Plant Physiology 177: 803-818.

Wang J., Tian C., Zhang C., Shi B., Cao X., Zhang T. Q., Zhao Z., Wang J. W., Jiao Y. (2017). Cytokinin signaling activates *WUSCHEL* expression during axillary meristem initiation. Plant Cell, 29: 1373-1387.

Wróblewska K., Bąbelewski P. (2010). The effect of benzyladenine and naphthalene acetic acid on rooting and subsequent growth of *Portulaca umbraticola* Kunth. Folia Horticulturae, 22: 39-44.

Yang W., Cortijo S., Korsbo N., Roszak P., Schiessl K., Gurzadyan A., Wightman R., Jönsson H., Meyerowitz E. (2021). Molecular mechanism of cytokinin-activated cell division in *Arabidopsis*. Science, 371: 1350-1355.

Zhang W., Fan J., Tan Q., Zhao M., Zhou T., Cao F. (2017). The effects of exogenous hormones on rooting process and the activities of key enzymes of *Malus hupehensis* stem cuttings. PLoS One, 12: Article e0172320.

Zhou X., Li R., Shen H., Yang L. (2023). Effect of exogenous plant growth regulators and rejuvenation measures on the endogenous hormone and enzyme activity responses of *Acer mono* maxim in cuttage rooting. International Journal of Molecular Sciences, 24: Article 11883.